Software Requirements Specification (SRS)

PORTKEY Food Delivery System

IEEE 830-1998 Format

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the PORTKEY Food Delivery System, an e-commerce platform that enables users to order food online from local restaurants, make payments, track deliveries, and interact with an intelligent chatbot for customer support.

1.2 Scope

The PORTKEY Food Delivery System provides customers with the convenience of ordering food from home or quickly identifying nearby restaurants for dine-in. The system allows users to browse local restaurants, order food, make payments (COD/Online), contact restaurants or delivery partners, and track deliveries in real-time. The system includes an ML-powered chatbot for 24x7 customer support and an admin dashboard for order management.

Key Benefits:
- Convenient food ordering from multiple restaurants
- Real-time order tracking and delivery updates
- Automated customer support through ML chatbot
- Secure payment processing
- Comprehensive admin dashboard for business management

1.3 Definitions, Acronyms, and Abbreviations

- UI - User Interface
- API - Application Programming Interface
- ML - Machine Learning
- COD - Cash on Delivery
- CRUD - Create, Read, Update, Delete
- LLM - Large Language Model
- REST - Representational State Transfer

- JSON - JavaScript Object Notation
- HTTPS - HyperText Transfer Protocol Secure
- SQL - Structured Query Language

1.4 References

- IEEE Std 830-1998 - IEEE Recommended Practice for Software Requirements Specifications
- ISO/IEC/IEEE 29148:2018 - Systems and software engineering requirements engineering
- W3C Web Standards for HTML5, CSS3, JavaScript

1.5 Overview

This document follows the IEEE 830-1998 standard structure for software requirements specifications. It provides a comprehensive description of the PORTKEY Food Delivery System, including functional and non-functional requirements, system interfaces, and design constraints.

2. Overall Description

2.1 Product Perspective

The PORTKEY Food Delivery System operates as a standalone web-based application that integrates with external services including:
- Payment gateways for secure transactions
- SMS/Email services for notifications
- Cloud storage for data synchronization
- Third-party mapping services for delivery tracking
- Restaurant management systems for menu updates

The system consists of three main components:
1. Customer Interface - Web-based frontend for food ordering
2. Admin Dashboard - Management interface for restaurants and orders
3. Backend Services - APIs, database, and ML chatbot engine

2.2 Product Functions

Primary Functions:
- User registration and authentication
- Restaurant browsing with menu display
- Shopping cart management
- Order placement and confirmation
- Payment processing (COD and online)
- Real-time order tracking
- Delivery partner assignment and communication
- ML chatbot for customer queries
- Admin dashboard for order and user management
- Push notifications for order updates

Secondary Functions:
- Order history and favorites
- Restaurant ratings and reviews
- Promotional offers and discounts

- Analytics and reporting
- Customer feedback collection

2.3 User Characteristics

Primary Users:
- Customers: General public with basic computer/smartphone literacy, familiar with e-commerce platforms
- Restaurant Owners: Business users comfortable with web applications for menu and order management
- Delivery Partners: Mobile-savvy individuals using the system for delivery coordination
- System Administrators: Technical users managing the overall system operations

User Expertise Levels:
- Novice users: Basic web navigation skills
- Intermediate users: Comfortable with online shopping and payments
- Expert users: Restaurant owners and admins with business application experience

2.4 Constraints

Technical Constraints:
- Must be compatible with modern web browsers (Chrome, Firefox, Safari, Edge)
- Responsive design supporting mobile devices (Android/iOS browsers)
- Internet connection required for all operations
- Minimum 8GB RAM for development environment
- Compatible with Windows development environment

Business Constraints:
- Compliance with local food delivery regulations
- Integration with existing restaurant POS systems
- Scalable architecture to handle peak ordering hours
- Data privacy compliance (GDPR, local regulations)

Security Constraints:
- Secure payment processing
- User data encryption and secure storage
- HTTPS protocol for all communications

2.5 Assumptions and Dependencies

Assumptions:
- Users have reliable internet connectivity
- Restaurants maintain updated menu information
- Delivery partners are available during operating hours
- Payment gateways maintain 99.9% uptime
- Users possess valid email addresses and phone numbers

Dependencies:
- Third-party payment gateway services
- Cloud hosting infrastructure reliability
- Restaurant cooperation for menu updates
- SMS/Email service provider availability
- Machine learning libraries for chatbot functionality

2.6 Apportioning of Requirements

Current Version Features:
- Core ordering and delivery functionality
- Basic chatbot with predefined responses
- Simple admin dashboard
- COD and basic online payment

Future Version Features:
- Advanced ML chatbot with natural language processing
- Mobile applications for Android and iOS
- Advanced analytics and business intelligence
- Integration with social media platforms
- Loyalty programs and gamification

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3. Specific Requirements

3.1 External Interfaces

3.1.1 User Interfaces
- Web Interface: Responsive HTML5/CSS3/JavaScript frontend
- Admin Dashboard: Comprehensive management interface
- Chatbot Interface: Floating chat widget with conversational UI

3.1.2 Hardware Interfaces
- Development Hardware: Intel i3/i5 processor, 8GB RAM minimum
- Storage: 500GB hard disk space
- Network: Stable broadband internet connection

3.1.3 Software Interfaces
- Frontend Framework: HTML5, CSS3, JavaScript
- Backend Framework: JavaScript
- Database: MySQL/MongoDB

- Operating System: Windows (development), Linux (production)

3.1.4 Communication Interfaces
- REST APIs: JSON-based web services
- HTTPS Protocol: Secure communication
- WebSocket: Real-time order tracking updates
- SMS Gateway: Order notifications
- Email Service: Registration and order confirmations

3.2 Functional Requirements

3.2.1 User Management

FR-1.1: The system shall allow new users to register with email, password, and contact information.
FR-1.2: The system shall authenticate users through secure login credentials.
FR-1.3: The system shall allow users to update their profile information.
FR-1.4: The system shall enable password reset functionality via email.

3.2.2 Restaurant Management

FR-2.1: The system shall display a list of available restaurants with basic information.
FR-2.2: The system shall show restaurant menus with item names, descriptions, and prices.
FR-2.3: The system shall allow filtering restaurants by cuisine type, rating, or delivery time.
FR-2.4: The system shall display restaurant contact information and operating hours.

3.2.3 Order Management

FR-3.1: The system shall allow users to add menu items to a shopping cart.
FR-3.2: The system shall enable users to modify cart contents (add, remove, update quantities).
FR-3.3: The system shall calculate total order amount including taxes and delivery charges.
FR-3.4: The system shall allow users to place orders with delivery address specification.
FR-3.5: The system shall generate unique order IDs for tracking purposes.

3.2.4 Payment Processing

FR-4.1: The system shall support Cash on Delivery (COD) payment method.
FR-4.2: The system shall integrate with online payment gateways for card payments.
FR-4.3: The system shall provide payment confirmation and receipt generation.
FR-4.4: The system shall handle payment failures gracefully with retry options.

3.2.5 Order Tracking

FR-5.1: The system shall provide real-time order status updates (preparing, dispatched, delivered).
FR-5.2: The system shall assign delivery partners to confirmed orders.
FR-5.3: The system shall allow communication between customers and delivery partners.
FR-5.4: The system shall send notifications for order status changes.

3.2.6 Chatbot Functionality

FR-6.1: The system shall provide an ML-powered chatbot for customer queries.
FR-6.2: The chatbot shall handle order inquiries, delivery status, and general FAQs.
FR-6.3: The chatbot shall escalate complex queries to human support.
FR-6.4: The system shall maintain chat history for user reference.

3.2.7 Admin Dashboard

FR-7.1: The system shall provide an admin interface for order management.
FR-7.2: The admin shall be able to view and manage user accounts.
FR-7.3: The system shall generate reports on orders, revenue, and user activity.
FR-7.4: The admin shall be able to manage restaurant information and menus.

3.3 Performance Requirements

PR-1: The system shall load the main page within 3 seconds on standard broadband connections.
PR-2: The system shall process order placement within 5 seconds of user confirmation.
PR-3: The system shall support concurrent access for up to 1000 simultaneous users.
PR-4: The chatbot shall respond to user queries within 2 seconds.
PR-5: The system shall maintain 99.5% uptime during operating hours.
PR-6: Database queries shall execute within 1 second for standard operations.
PR-7: The system shall handle peak traffic during meal times without degradation.

3.4 Logical Database Requirements

3.4.1 Data Storage
The database shall store the following information:
- User Data: UserID, Name, Email, Password (encrypted), Phone, Address, Role
- Restaurant Data: RestaurantID, Name, Address, Contact, Operating Hours, Cuisine Type
- Menu Data: ItemID, RestaurantID, Name, Description, Price, Category, Availability
- Order Data: OrderID, UserID, RestaurantID, Items, TotalAmount, Status, Timestamp
- Delivery Data: PartnerID, Name, Phone, Status, Current Orders

3.4.2 Data Relationships
- One User can place many Orders (1:N)
- One Restaurant has many Menu Items (1:N)
- One Order contains multiple Order Items (1:N)
- One Delivery Partner handles one Order at a time (1:1)

3.4.3 Data Integrity
- Primary keys for all entities
- Foreign key constraints for referential integrity
- Data validation for email formats and phone numbers
- Encryption for sensitive information (passwords, payment data)

3.5 Design Constraints

DC-1: The system shall use responsive web design principles for mobile compatibility.
DC-2: The frontend shall be developed using HTML5, CSS3, and JavaScript.
DC-3: The backend shall use JavaScript..
DC-4: The database shall be MySQL or MongoDB.
DC-5: The system shall follow RESTful API design principles.
DC-6: All communications shall use HTTPS protocol for security.
DC-7: The chatbot shall be implemented using Python-based ML libraries.

3.6 Software System Attributes

3.6.1 Reliability
- System availability of 99.5% during operating hours
- Automatic error recovery mechanisms
- Data backup and recovery procedures
- Graceful handling of system failures

3.6.2 Security
- User authentication and authorization
- Encrypted password storage
- Secure payment processing
- Data transmission encryption (HTTPS)
- Input validation to prevent SQL injection

3.6.3 Maintainability
- Modular code architecture
- Comprehensive documentation
- Version control using Git/GitHub
- Automated testing procedures
- Clear separation of frontend and backend components

3.6.4 Portability
- Cross-browser compatibility
- Responsive design for various screen sizes
- Database abstraction for multiple database systems
- Cloud deployment capability

3.6.5 Usability
- Intuitive user interface design
- Clear navigation and menu structure
- Accessibility features for disabled users
- Multi-language support capability
- Help documentation and tooltips

3.7 Organization of Specific Requirements

Requirements are organized into the following modules:

User Management Module:
- Registration, login, profile management
- Authentication and authorization

Restaurant Module:
- Restaurant listing, menu display
- Search and filtering capabilities

Order Module:
- Cart management, order placement
- Order confirmation and tracking

Payment Module:
- Payment gateway integration
- Transaction processing and receipts

Delivery Module:
- Delivery partner assignment
- Real-time tracking and communication

Chatbot Module:
- ML-based query processing
- Customer support automation

Admin Module:
- Dashboard for system management
- Reporting and analytics

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Appendices

Appendix A: Glossary

Cart: Virtual shopping basket containing selected menu items before order placement.
Chatbot: Automated conversational agent using machine learning for customer support.
Delivery Partner: Individual responsible for delivering orders from restaurant to customer.

Order Status: Current state of an order (placed, preparing, dispatched, delivered, cancelled).
Real-time Tracking: Live updates on order preparation and delivery progress.
Restaurant Listing: Catalog of available restaurants with menu and contact information.

Appendix B: Use Case Diagrams

Use case diagrams and system architecture diagrams are maintained in separate design documents.

Appendix C: Revision History

Version	Date	Description	Author
1.0	Sept 2025	Initial SRS Document	Team PORTKEY

Appendix D: Validation Criteria

- User acceptance testing with target customers
- Performance testing under simulated load
- Security testing for payment processing
- Compatibility testing across browsers and devices
- Chatbot accuracy testing with sample queries

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